

MLMC Seminar

Efficient White Noise Sampling and Coupling for Multilevel Monte Carlo with Nonnested Meshes

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Abstract

- In many applications it is required to generate samples from specific points of a Gaussian random field.
- This produces a Gaussian random vector with specific distribution parameters, namely the mean vector and the covariance matrix.
- Algorithms like Box-Muller let us easily sample from a standard normal distribution.
- While the desired mean can be achieved via the summation of the mean vector, the right covariance is quite more involved since it is necessary an expensive **Cholesky factorization**.

- In the particular case presented by this article the covariance matrix is the **mass matrix of a finite element method**
- The main idea to make the sampling more efficient is to exploit the assembling of the global mass matrix as a linear combination of all mass matrices associated to each element of the mesh.
- Due to linearity it is possible to factorize the smaller mass matrices of the elements instead of the more extended global one.

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Problem Setting

Given the following elliptic PDE with random coefficient u :

$$\begin{cases} -\nabla \cdot (e^{u(x,\omega)} \nabla q(x, \omega)) = 1, & x \in G = [-0.5, 0.5]^d, \quad \omega \in \Omega \\ q(x, \omega) = 0, & x \in \partial G, \quad \omega \in \Omega \end{cases} \quad (1)$$

We are interested in estimating the expected value of the L^2 -norm of the solution q :

$$\mathbb{E}[\|q\|_{L^2(G)}] = \int_{\Omega} \|q\|_{L^2(G)}(\omega) d\mathbb{P}(\omega) \quad (2)$$

This task can be also seen as computing the expected value of a real-valued random variable

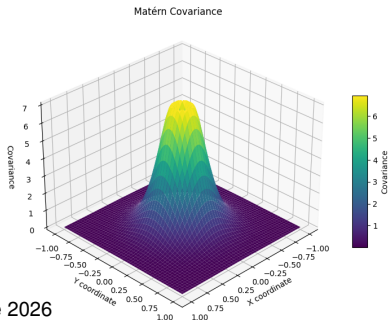
$$\mathcal{P} := \|q\|_{L^2(G)} : \Omega \rightarrow \mathbb{R}.$$

Matérn Covariance

Assume u is a Gaussian random field with zero mean and Matérn covariance $\mathcal{C}$:

$$\mathcal{C}(x, y) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} (\kappa r)^\nu K_\nu(\kappa r), \quad r = \|x - y\|_2, \quad \kappa = \frac{\sqrt{8\nu}}{\lambda}, \quad x, y \in D \quad (3)$$

where $\sigma^2, \nu, \lambda > 0$ are respectively the variance, smoothness and correlation length while K_ν is the modified Bessel function of second kind.



Discrete Sampling

In practice, samples of u are needed only at discrete locations $x_1, \dots, x_m \in D$.

A simple sampling strategy consists of drawing realisations of a Gaussian vector V in $\mathbb{R}^m$

$$u \sim \mathcal{N}(0, C) \text{ with } u_i = u(x_i) \text{ and covariance matrix } C_{ij} = \mathbb{E}[u(x_i) u(x_j)] \in \mathbb{R}^{m \times m}.$$

Then using Box-Muller to sample $\mathbf{z} \sim \mathcal{N}(0, I)$, $I \in \mathbb{R}^{n \times n}$, and the factorization $C = HH^T$, $H \in \mathbb{R}^{m \times n}$, gives $u = Hz$ since:

$$\mathbb{E}[uu^T] = \mathbb{E}[Hz(Hz)^T] = H \mathbb{E}[zz^T] H^T = H I H^T = C \quad (4)$$

This approach requires the factorization of a dense covariance matrix: complexity $\mathcal{O}(m^3)$.

An alternative Sampling Method: the SPDE Approach

Among all methods for the generation of random field samples we choose a particular one which exploits both the use of the generated sample as part of a finite element method, and its specific covariance function.

SPDE characterisation of Matérn fields

A standard result in SPDE theory shows that a Gaussian random field with Matérn covariance satisfies the following elliptic SPDE:

$$(I - \kappa^{-2}\Delta)^k u(x, \omega) = \dot{W}(x, \omega), \quad x \in \mathbb{R}^d, \quad \omega \in \Omega, \quad \nu = 2k - d/2 > 0 \quad (5)$$

where $\dot{W}$ is Gaussian white noise defined on $\mathbb{R}^d$.

- It is not feasible to solve the PDE on all of $\mathbb{R}^d$, so we restrict to a domain D such that $G \subset D$.
- This yields to a choice of the boundary value on ∂D but if $\text{dist}(\partial G, \partial D) \gg \lambda$ the truncation error is negligible as the specific type condition that we choose.
- For simplicity we consider the case $k = 1$ with homogeneous Dirichlet boundary conditions.

White Noise: Definition

Definition 1 (White Noise)

Let $D \subset \mathbb{R}^d$ be an open domain. $\dot{W} \in \mathcal{L}(L^2(D), L^2(\Omega, \mathbb{R}))$, i.e. a linear functional on $L^2(D)$ inducing square-integrable real-valued random variables, such that for any collection $\{\varphi_i\} \subset L^2(D)$, setting $b_i = \langle \dot{W}, \varphi_i \rangle$, the $\{b_i\}$ are jointly Gaussian with zero mean and covariance:

$$\begin{aligned}\mathbb{E}[b_i b_j] &= \mathbb{E}\left[\int_D \varphi_i(x) \dot{W}(x) dx \int_D \varphi_j(y) \dot{W}(y) dy\right] \\ &= \int_{D \times D} \varphi_i(x) \varphi_j(y) \underbrace{\mathbb{E}[\dot{W}(x) \dot{W}(y)]}_{:= \delta(x-y)} dx dy = (\varphi_i, \varphi_j)\end{aligned}\tag{6}$$

FEM Discretisation of the SPDE

This sampling technique gives a random field sample with the right covariance function but we have to solve a SPDE.

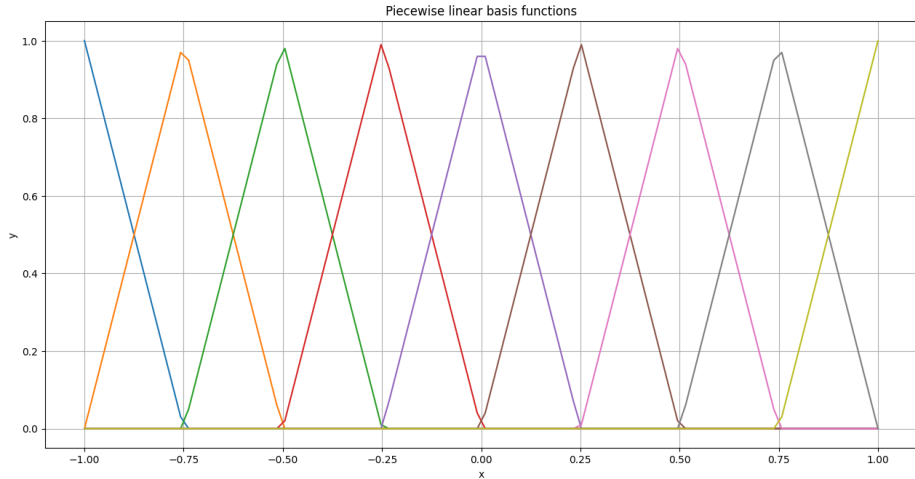
A clever way of doing that is choose as FEM space an extension of FEM space that we use on G to solve the original PDE with random coefficient, or vice versa, choose the FEM on G as a restriction of the one define on D .

Doing so we obtain for free the value of the random field on the space spacial nodes where we will solve the original problem.

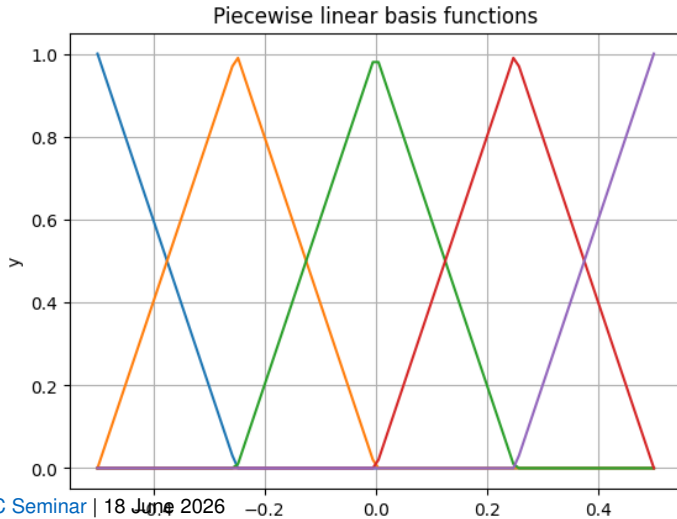
$V_h^D = \text{span}\{\varphi_1, \dots, \varphi_m\} \subset H_0^1(D)$ (e.g. linear Lagrange basis functions) such that

$$V_h^G = \{\varphi_i \in V_h^D \mid \text{supp}(\varphi_i) \subset G\}, \quad |V_h^G| = m', \quad \subset V_h^D, \quad V_h^G \subset H_0^1(G) \cap H_0^1(D). \quad (7)$$

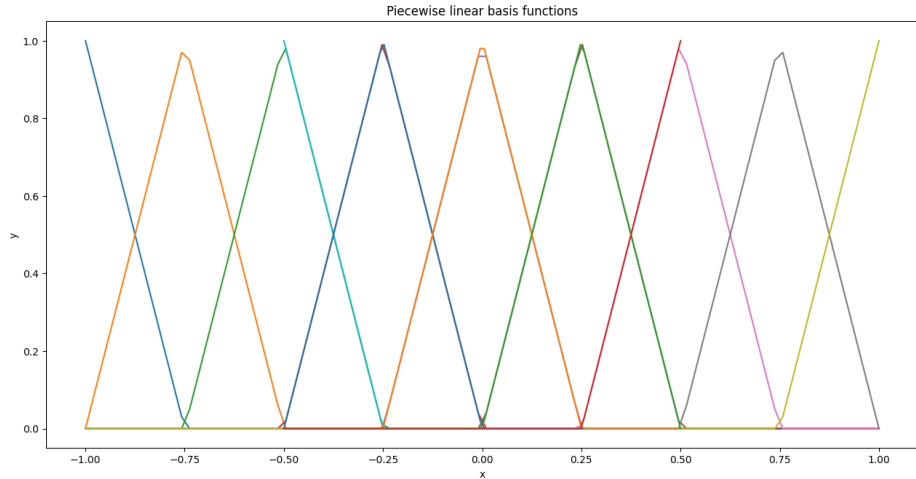
1d Example of V_h^D



1d Example of V_h^G



1d Example of $V_h^G \subset V_h^D$



The discretised weak form of the PSD reads:

$$(u_h, v_h) + \kappa^{-2}(\nabla u_h, \nabla v_h) = \langle \dot{W}, v_h \rangle \quad \forall v_h \in V_h^D. \quad (8)$$

The coefficients $\mathbf{u}_i$ such that $u_h = \sum_{i=1}^m \mathbf{u}_i \varphi_i$ satisfy $A\mathbf{u} = b$ with:

$$A_{ij} = (\varphi_i, \varphi_j) + \kappa^{-2}(\nabla \varphi_i, \nabla \varphi_j), \quad b_i = \langle \dot{W}, \varphi_i \rangle, \quad i = 1, \dots, m. \quad (9)$$

The discretised right-hand side is Gaussian by definition of white noise:

$$b \sim \mathcal{N}(0, M), \quad M_{ij} = (\varphi_i, \varphi_j). \quad (10)$$

Thus, sampling white noise reduces to sampling a Gaussian vector with **mass matrix covariance**. This alone does not yet improve the naive approach since an expensive global Cholesky of M would still be needed.

From Global to Local: Element-wise Mass Matrix

Key observation: M assembles as a linear combination of element-wise matrices.

For a mesh of n elements with m_e degrees of freedom per element:

$$M = L^T \text{diag}(M_e) L \quad (11)$$

where $\text{diag}(M_e)$ is block-diagonal of size $nm_e \times nm_e$, and L is a Boolean assembling matrix of size $nm_e \times m$ encoding the local-to-global map.

We can factorise each local mass matrix $M_e = H_e H_e^T$ and then obtain the global Cholesky factorization as

$$M = L^T \text{diag}(H_e H_e^T) L = (L^T \text{diag}(H_e)) (L^T \text{diag}(H_e))^T = H H^T. \quad (12)$$

With $H := L^T \text{diag}(H_e)$ we can sample $b = H \mathbf{z}$, $\mathbf{z} \sim \mathcal{N}(0, I)$, $\mathbf{z} \in \mathbb{R}^{m_e n}$, since:

$$\mathbb{E}[bb^T] = H \mathbb{E}[\mathbf{z}\mathbf{z}^T] H^T = M \quad (13)$$

Local Sampling: Algorithm and Cost

Equivalently, we can adopt a **local sampling strategy** on each element e .

Let's consider $\mathbf{z}_e \sim \mathcal{N}(0, I)$ of length m_e .

Then:

$$\mathbf{b} = H\mathbf{z} = \sum_{e=1}^n L_e^T H_e \mathbf{z}_e = \sum_{e=1}^n L_e^T \mathbf{b}_e, \quad \mathbf{b}_e \sim \mathcal{N}(0, M_e). \quad (14)$$

The global sampling problem reduces to n **independent local sampling problems** which can be immediately parallelized.

In both cases the computational cost is reduced from $\mathcal{O}(m^3)$ to $\mathcal{O}(nm_e^3)$, $m_e \ll m$.

Moreover, in the special case of affine linear reference element:

$$M_e/|e| = \text{const} \quad \forall e$$

Hence only one local Cholesky factorization is needed which drastically reduces the computational cost to $\mathcal{O}(m_e^3)$.

Solving Both PDEs and the Monte Carlo Estimator

So far we have developed an efficient way to sample a white noise realization $b(\omega)$ but to generate a single Matérn random field sample we still have to:

1. solve the SPDE: $Au_h = b(\omega)$ on D_h to obtain $u_h(\cdot, \omega)$.
2. solve the elliptic PDE with random coefficient: $\tilde{A}(\omega) q_h = \tilde{b}$ on G_h , with associated discrete variational formulation

$$\tilde{A}_{ij} = (e^{u_h(\cdot, \omega)} \nabla \varphi_i, \nabla \varphi_j), \quad \tilde{b}_i = (1, \varphi_i), \quad i = 1, \dots, m' \quad \text{and} \quad \varphi_i \in V_h^G.$$

Assuming $m \approx m'$, both PDE can be solved at a cost of $\mathcal{O}(m)$ per sample using an optimal preconditioner.

The standard unbiased Monte Carlo estimator is:

$$E^{\text{MC}}[\mathcal{P}] \approx E^{\text{MC}}[\mathcal{P}_h] = \frac{1}{N} \sum_{i=1}^N \|q_h(\cdot, \omega_i)\|_{L^2(G)}^2 = \frac{1}{N} \sum_{i=1}^N \sqrt{q_h(\omega_i)^T M q_h(\omega_i)}. \quad (15)$$

This has **slow variance convergence**, making standard MC too expensive for fine meshes.

Multilevel Monte Carlo

The standard solution is to use Multilevel Monte Carlo.

Consider a sequence of meshes with decreasing element diameters $\{h_l\}_{l=1}^L$.

The MLMC estimator is:

$$E^{\text{MLMC}}[\mathcal{P}_{h_L}] = \sum_{l=1}^L \frac{1}{N_l} \sum_{n=1}^{N_l} \left(\|q_{h_l}(\omega_l^n)\|_{L^2(G)} - \|q_{h_{l-1}}(\omega_l^n)\|_{L^2(G)} \right) \quad (16)$$

The efficiency of this method relies on balancing cost and variance across levels:

- **Coarse levels** (l small): large variance of correction terms, many samples needed — but each sample is cheap (coarse mesh).
- **Fine levels** (l large): high cost per sample — but few samples needed because variance of the correction $Y_l = \|q_{h_l}\| - \|q_{h_{l-1}}\|$ is small.

Variance Reduction in Multilevel Monte Carlo

We provide an intuitive explanation of the variance reduction mechanism.

Let

$$Q_l := \|q_{h_l}(\omega)\|_{L^2(G)}, \quad Y_l := Q_l - Q_{l-1}.$$

Then

$$\text{Var}(Y_l) = \text{Var}(Q_l) + \text{Var}(Q_{l-1}) - 2 \text{Cov}(Q_l, Q_{l-1}).$$

The quantities Q_l and Q_{l-1} are computed using the same realization ω . Since h_l and h_{l-1} are consecutive mesh levels, the corresponding FEM approximations are similar:

$$u_{h_l}(\cdot, \omega) \approx u_{h_{l-1}}(\cdot, \omega).$$

Assuming the quantity of interest depends continuously on the FEM solution,

$$q_{h_l}(\cdot, \omega) \approx q_{h_{l-1}}(\cdot, \omega),$$

and therefore

$$Q_l \approx Q_{l-1}.$$

Hence Q_l and Q_{l-1} are strongly positively correlated:

$$\text{Corr}(Q_l, Q_{l-1}) = \frac{\text{Cov}(Q_l, Q_{l-1})}{\sqrt{\text{Var}(Q_l)\text{Var}(Q_{l-1})}} \approx 1,$$

which implies

$$\text{Cov}(Q_l, Q_{l-1}) \approx \sqrt{\text{Var}(Q_l)\text{Var}(Q_{l-1})}.$$

Substituting this approximation into the variance formula,

$$\text{Var}(Y_l) \approx \left(\sqrt{\text{Var}(Q_l)} - \sqrt{\text{Var}(Q_{l-1})} \right)^2 \ll \text{Var}(Q_l).$$

Coupled Sampling Across Levels

The key to variance reduction is using the **same white noise realisation** on both levels l and $l - 1$. This means solving the SPDE on two different meshes for the same ω_l^n .

Let $V_{h_l}^D = \text{span}(\varphi_1^l, \dots, \varphi_{m_l}^l)$ and $V_{h_{l-1}}^D = \text{span}(\varphi_1^{l-1}, \dots, \varphi_{m_{l-1}}^{l-1})$.

The two discretised variational problems coupled by a common white noise sample are:

$$\begin{aligned} (u_{h_l}, v_{h_l}) + \kappa^{-2}(\nabla u_{h_l}, \nabla v_{h_l}) &= \langle \dot{W}, v_{h_l} \rangle (\omega_l^n) \quad \forall v_{h_l} \in V_{h_l}^D \\ (u_{h_{l-1}}, v_{h_{l-1}}) + \kappa^{-2}(\nabla u_{h_{l-1}}, \nabla v_{h_{l-1}}) &= \langle \dot{W}, v_{h_{l-1}} \rangle (\omega_l^n) \quad \forall v_{h_{l-1}} \in V_{h_{l-1}}^D \end{aligned} \quad (17)$$

where the right-hand sides are coupled by White Noise definition:

$$\mathbb{E}[\langle \dot{W}, v^l \rangle \langle \dot{W}, v^{l-1} \rangle] = (v^l, v^{l-1}).$$

The unknowns satisfy the block-diagonal linear system:

$$\begin{pmatrix} A_l & 0 \\ 0 & A_{l-1} \end{pmatrix} \begin{pmatrix} \mathbf{u}_l \\ \mathbf{u}_{l-1} \end{pmatrix} = \begin{pmatrix} b_l \\ b_{l-1} \end{pmatrix} \quad (18)$$

The Coupled Mass Matrix

The white noise sample vector $b \in \mathbb{R}^{m_l+m_{l-1}}$ satisfies:

$$b \sim \mathcal{N}(0, M), \quad M = \begin{pmatrix} M_l & M_{l,l-1} \\ (M_{l,l-1})^T & M_{l-1} \end{pmatrix} \quad (19)$$

where:

$$(M_{l,l-1})_{ij} = (\varphi_i^l, \varphi_j^{l-1}), \quad M_{ij} = (\varphi_i^l, \varphi_j^l), \quad M_{l,l-1} \in \mathbb{R}^{m_l \times m_{l-1}}$$

Challenge: if the meshes are **non-nested**, there is no direct linear combination that assembles M element-wise, because $M_{l,l-1}$ involves inner products of basis functions defined on *different* meshes.

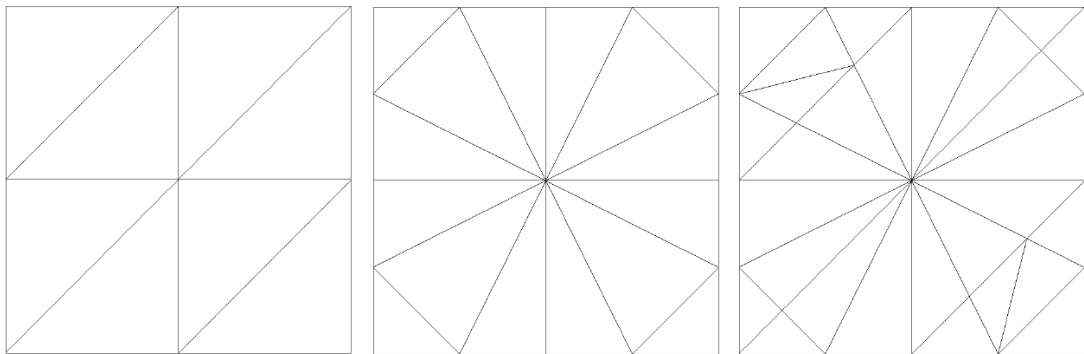
Using independent samples (off-diagonal blocks zero) would destroy the coupling and prevent variance reduction.

⇒ Solution: construct a **super-mesh** will simplify the computation of inner product across different meshes.

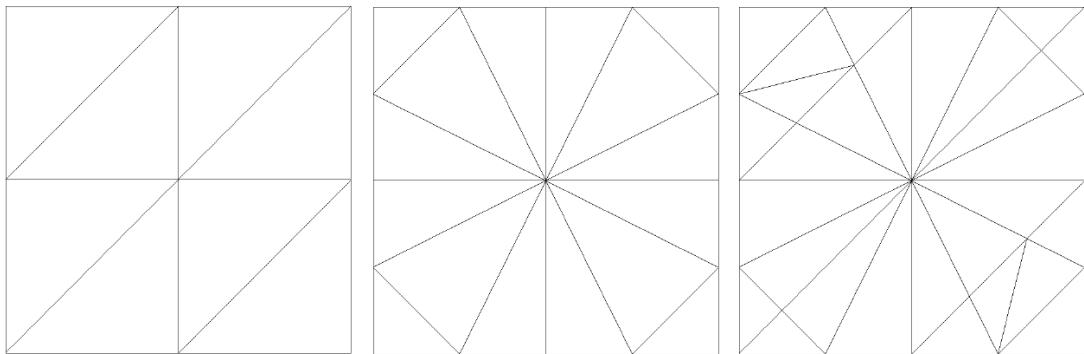
Definition 2 (Super-mesh)

Let D be a compact subset of $\mathbb{R}^d$, and let $\mathcal{T}_a, \mathcal{T}_b$ be two tessellations of D . A super-mesh $\mathcal{S}$ of $\mathcal{T}_a$ and $\mathcal{T}_b$ is a common refinement of both meshes, i.e. a triangulation of D such that:

1. $\text{vertices}(\mathcal{T}_a) \cup \text{vertices}(\mathcal{T}_b) \subseteq \text{vertices}(\mathcal{S})$
2. $\text{vol}(e_{\mathcal{S}} \cap e) \in \{0, \text{vol}(e_{\mathcal{S}})\} \quad \forall e_{\mathcal{S}} \in \mathcal{S}, e \in (\mathcal{T}_a \cup \mathcal{T}_b)$



- Each super-mesh element is formed by the intersection of **exactly one pair** of parent elements $e^l \in D_{h_l}$ and $e^{l-1} \in D_{h_{l-1}}$.



- Let m_{e^l} and $m_{e^{l-1}}$ denote the number of DOFs of $V_{h_l}^D$ and $V_{h_{l-1}}^D$ restricted to those elements. Only inner products between these $m_e = m_{e^l} + m_{e^{l-1}}$ basis functions are non-zero.

Super-mesh Local Mass Matrix and Global Assembly

The local mass matrix over a super-mesh element e is:

$$M_e = \begin{pmatrix} M_e^l & M_e^{l,l-1} \\ (M_e^{l,l-1})^T & M_e^{l-1} \end{pmatrix} \in \mathbb{R}^{m_e \times m_e}, \quad (M_e^{l,l-1})_{ij} = \int_e \varphi_i^l \varphi_j^{l-1} dx \quad (20)$$

Using Boolean assembling matrices L^l and L^{l-1} of the two original meshes:

$$M = \begin{pmatrix} (L^l)^T & (L^{l-1})^T \end{pmatrix} \begin{pmatrix} \text{diag}(M_e^l) & \text{diag}(M_e^{l,l-1}) \\ \text{diag}(M_e^{l,l-1})^T & \text{diag}(M_e^{l-1}) \end{pmatrix} \begin{pmatrix} L^l \\ L^{l-1} \end{pmatrix} = \begin{pmatrix} M_l & M_{l,l-1} \\ (M_{l,l-1})^T & M_{l-1} \end{pmatrix} \quad (21)$$

This gives the desired linear assembling of the coupled mass matrix and the **same efficient local sampling technique** now applies. For each super-mesh element e , draw $b_e \sim \mathcal{N}(0, M_e)$ and split $b_e = ((b_e^l)^T (b_e^{l-1})^T)^T$. By construction:

$$b_e^l \sim \mathcal{N}(0, M_e^l), \quad b_e^{l-1} \sim \mathcal{N}(0, M_e^{l-1}), \quad \mathbb{E}[b_e^l (b_e^{l-1})^T] = M_e^{l,l-1} \quad (22)$$

Assembling the Coupled Sample

Using the Boolean matrices, for each super-mesh element e :

$$b^l = \sum_{e=1}^n (L_e^l)^T b_e^l, \quad b^{l-1} = \sum_{e=1}^n (L_e^{l-1})^T b_e^{l-1} \quad (23)$$

where n is the number of super-mesh elements.

This enforces the correct distribution since sums of Gaussian random variables are Gaussian. In particular:

$$\mathbb{E}[b^l (b^l)^T] = \sum_{i,j} (L_i^l)^T \mathbb{E}[b_i^l (b_j^l)^T] L_j^l = \sum_i (L_i^l)^T M_i^l L_i^l = (L^l)^T \text{diag}(M_i^l) L^l = M_l \quad (24)$$

and analogously for $l - 1$ and the mixed term giving $M_{l,l-1}$.

Lemma: Affine Reference Element

Lemma 1

Let $V_{h_l}^D, V_{h_{l-1}}^D$ be two FEM spaces over tessellations $D_{h_l}, D_{h_{l-1}}$ of D . Let $\mathcal{S}$ be a super-mesh, $e \in \mathcal{S}$. Assume $V_{h_{l-1}}^D|_e \subseteq V_{h_l}^D|_e$ (nested restrictions) and M_e^l non-singular. Then:

■ $\text{rank}(M_e) = \text{rank}(M_e^l)$ and $M_e^{l-1} = (M_e^{l,l-1})^T (M_e^l)^{-1} M_e^{l,l-1}$

If the transformation to the reference element is affine and $V^S|_e$ is a FEM subspace over $\mathcal{S}$, with $V_{h_{l-1}}^D|_e, V_{h_l}^D|_e \subseteq V^S|_e$, then there exist local interpolation matrices R_e^l and R_e^{l-1} such that:

■

$$M_e^l = (R_e^l)^T M_e^S R_e^l, \quad M_e^{l-1} = (R_e^{l-1})^T M_e^S R_e^{l-1}, \quad M_e^{l,l-1} = (R_e^l)^T M_e^S R_e^{l-1} \quad (25)$$

In practice for Lagrange FEM:

$$b_e^k = (R_e^k)^T H_r |e_r|^{-1/2} |e|^{1/2} \mathbf{z}_e, \quad \mathbf{z}_e \sim \mathcal{N}(0, I), \quad k \in \{l-1, l\} \quad (26)$$

where $M_e^S / |e| = H_r H_r^T / |e_r| = \text{const}$ is the Cholesky factorisation over the reference element e_r . H_r is computed **only once**.

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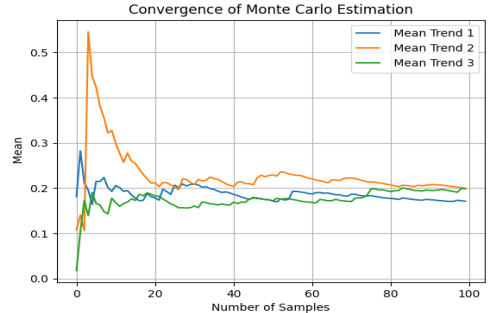
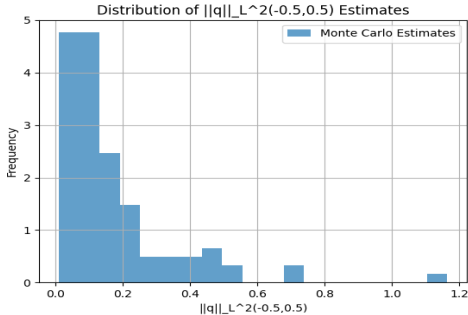
1D Numerical Experiments

We aim to replicate the results shown in the article in the **simpler 1D case**. This allows us to focus on the main ideas rather than the more complex FEM reasoning which is beyond the scope of the seminar.

Setup:

- Domains: $G = [-0.5, 0.5]$ and $D = [-1, 1]$
- FEM function space: piecewise linear test function (Lemma holds).
- Matérn covariance with $\kappa = 10$.
- Quantity of interest: $\mathcal{P} \approx \|q_h\|_{L^2(G)}$, estimated via Monte Carlo.
- $N = 100$, $N = 1\,000$ and $N = 10\,000$
-  [mlmc_fem_1d_solver](#)

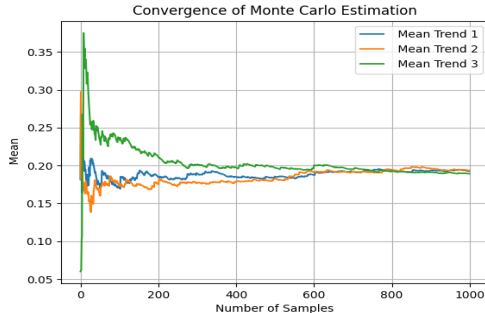
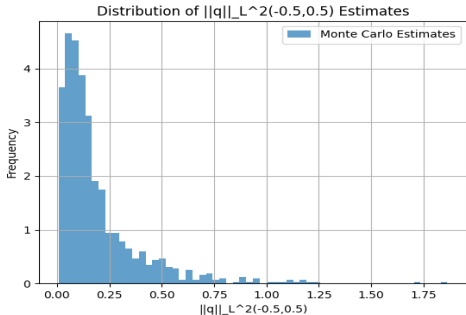
1D Results: Distribution and Convergence - 100 samples



Estimated expected value of $\|q\|_L^2(-0.5,0.5)$ using Monte Carlo with 100 samples, seed 2026: 0.17095288589067742.

Time taken for Monte Carlo estimation: 0.0 hours 0 min 8.39 seconds.

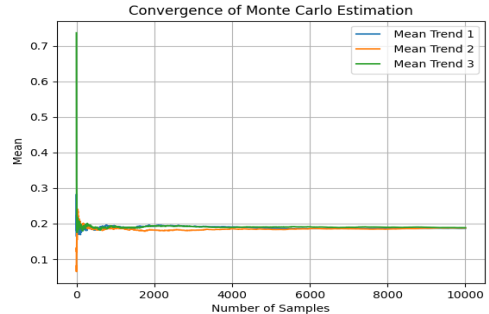
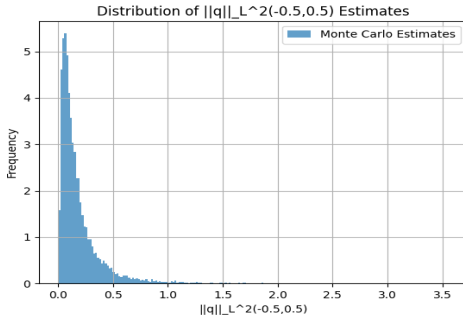
1D Results: Distribution and Convergence - 1000 samples



Estimated expected value of $\|q\|_{L^2(-0.5,0.5)}$ using Monte Carlo with 1000 samples, seed 2026: 0.19263251472509055.

Time taken for Monte Carlo estimation: 0.0 hours 1 min 18.32 seconds.

1D Results: Distribution and Convergence - 10000 samples



Estimated expected value of $||q||_L^2(-0.5,0.5)$ using Monte Carlo with 10000 samples, seed 2026: 0.18688853989513146.

Time taken for Monte Carlo estimation: 0.0 hours 8 min 14.11 seconds.

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Discussion

Main advantage:

- The local, element-wise structure of the sampling procedure leads to a **straightforwardly parallelisable** implementation.

Main drawback:

- Generating each sample for both Monte Carlo and Multilevel Monte Carlo requires **solving two PDEs** to obtain a single realisation for estimating the desired quantity.

Summary

- Sampling a Matérn random field via the SPDE approach reduces to sampling a Gaussian vector with **mass matrix covariance**, avoiding dense covariance factorisation ($\mathcal{O}(m^3) \rightarrow \mathcal{O}(m_e^3)$).
- The global mass matrix assembles from local element matrices, enabling **element-wise Cholesky factorisation** and embarrassingly parallel sampling; for Lagrange elements a single reference factorisation suffices.
- MLMC variance reduction requires **coupled sampling** of u_{h_l} and $u_{h_{l-1}}$ from the same white noise realisation; for non-nested meshes this is achieved via a **super-mesh** and local interpolation matrices R_e^l, R_e^{l-1} .
- The method is effective but overall efficiency depends on available **parallel hardware**, since each sample requires solving two PDEs.